DATA PROVENANCE, USAGE, PROTECTION, AND RELEASE RECORD

Record field	Details
Letter Number	010/DRPM/Univ-BD/IX/2026
Research title	Evaluating Late Fusion under Fragmented Multimodal Evidence for Indonesian Cyberbullying Detection
Principal researcher	Assoc. Prof. Dr. Ari Muzakir, S.Kom., M.Cs
Academic unit	S1 Informatics Engineering Study Program, Faculty of Science and Technology, Universitas Bina Darma
Document purpose	To document data provenance, research use, de-identification, protection measures, and public-release boundaries associated with the reported experiments and repository artefacts
Associated public repository	https://github.com/arimuzakir/Evaluating-late-fusion-indonesian-cyberbullying
Date	September 23, 2026

1. Purpose and Scope

This document records the origin, transformation, use, protection, and public-release status of the research data associated with the manuscript entitled "Evaluating Late Fusion under Fragmented Multimodal Evidence for Indonesian Cyberbullying Detection." It is prepared as formal administrative documentation to support scientific accountability, journal clarification, and repository governance.

The record covers the composition of the research corpus, the documented acquisition basis, annotation and audit procedures, experimental use, de-identification measures, release controls, and reproducibility boundaries. This document is an administrative research record. It is not an institutional ethics approval, an ethics-exemption determination, or a blanket permission to redistribute third-party platform content.

2. Corpus Origin and Dataset Composition

The study analysed an Indonesian social-media corpus comprising two connected experimental populations: standalone comments and temporally segmented video observations. According to the research record, social-media materials were collected through official platform APIs and subsequently processed to remove direct user-identifying information before public release. Archived benchmark resources and derived research artefacts, where applicable, were handled as secondary research materials within the documented pipeline.

Population / stage	Observations	Documented description
Original labelled corpus	13,307	8,906 standalone comments and 4,401 video windows from 233 source videos
Standalone comments, initial split	8,906	4,646 positive and 4,260 negative comments
Comments after agreement filter	8,216	Observations with a narrow 3–2 majority were excluded from the standalone-comment benchmark
Eligible comments	8,196	After text normalization and duplicate checks; 4,397 positive and 3,799 negative

Comment benchmark test set	1,226	Held-out comments used for the standalone benchmark reported in the manuscript
Aligned video evaluation cohort	3,951	2,326 positive and 1,625 negative windows from 214 linked source groups
Alignment exclusions	450	Source windows excluded after alignment checks on segment identifiers, keyframe links, and normalized linguistic content
Video audit sample	100	Blinded audit sample from 68 source groups

3. Acquisition Basis and Provenance Statement

The research team states that source data were acquired from public social-media services through official platform APIs. The platforms represented in the corpus were Facebook, Instagram, TikTok, YouTube, and X. The provenance model distinguishes collection from live platforms, transformation into research-ready records, and release of reviewed derived artefacts.

For each platform, the research team maintained internal records describing the relevant source, the data-access route used during collection, and the subsequent transformations applied before model development and public release. Direct identifiers such as account names, profile links, user identifiers, and comparable fields were removed or withheld from the released research artefacts. Original audiovisual materials and any restricted source mappings were not included in the public repository.

Provenance component	**Recorded status**
Platform sources	Facebook, Instagram, TikTok, YouTube, and X
Collection basis	Official platform APIs, according to the research team record
Unit of analysis	Standalone comments and 10-second video windows with linked multimodal evidence
Derived modalities	Keyframes, 16-dimensional acoustic features, OCR text, STT text, labels, stored predictions, and summary statistics
Public release basis	Reviewed de-identified artefacts and derived outputs only
Restricted materials	Original videos, source audio, identifiable frames, unreduced source text, identity mappings, and non-public credentials

4. Data Processing, Annotation, and Audit Trail

Video source material was segmented into 10-second windows with a 3-second stride. The principal modalities used in the multimodal experiments were a visual keyframe, a 16-dimensional acoustic representation, and OCR/STT-derived text aligned to the same time window. Original videos and derived modalities were not counted as separate observations. For the text branch, preprocessing included HTML-entity decoding, placeholder replacement for URLs and mentions, emoji-to-text conversion, normalization of informal language, preservation of hashtag words, and reduction of repeated characters and whitespace.

Annotation followed a five-vote procedure for each observation, producing an overall Fleiss' kappa of 0.8853 for the main vote matrix. Historical expert review records covered 1,331 observations. For the updated v1.1 reference set, four expert-adjudicated video-window label changes were retained. The manuscript further reports a blinded audit of 100 video windows drawn from 68 source groups. Pre-consensus aggression agreement in this audit was 95.0% (Cohen's kappa = 0.8998). After adjudication, all sampled labels matched the reference labels, comprising 45 negative and 55 positive windows.

Within the audited sample, faithful OCR outputs were identified in 43 of 52 available extractions (82.7%), and faithful STT outputs were identified in 51 of 64 available outputs (79.7%). Table 5 of the manuscript separately reports inter-rater agreement for OCR and STT fidelity as 48/52 (kappa = 0.8514) and 60/64 (kappa = 0.8929), respectively. Among the 55 positive audited windows, item-level keyframe relevance received score 2 in 46 cases and score 1 in nine cases.

5. Research Use in the Reported Experiments

The corpus supported two distinct experimental tracks reported in the manuscript. The first was a standalone-comment benchmark using lexical and contextual text models on held-out comments. The second was the principal multimodal video evaluation using source-disjoint partitions and aligned windows. These populations were analysed separately because their selection procedures and input characteristics differed.

Experimental use	Documented scope and results
Standalone-comment benchmark	1,226 held-out eligible comments; contextual language models, TF-IDF baseline, and ensemble comparison; best Macro F1 = 0.9663
Multimodal video evaluation	3,951 aligned windows from 214 linked source groups; 21 model configurations using visual, acoustic, and text evidence
Primary fusion comparison	Equal VAT Macro F1 = 0.5670 versus extracted-text Macro F1 = 0.5703 at seed 42; paired difference = -0.0034, 95% CI [-0.0179, 0.0116], p = 0.6608
Three-seed stability	Extracted text = 0.5591 ± 0.0111; Equal VAT = 0.5590 ± 0.0084
Audit and diagnostics	100-window audit, threshold analyses, calibration diagnostics, source-group bootstrap, permutation testing, and frozen-feature cross-platform probes

6. Data Protection and De-identification Measures

The public research release was limited to de-identified or pseudonymised research artefacts. Direct personal identifiers were removed during processing. The release design further restricted source information that could directly expose content creators, account owners, or source-post links. Public artefacts were limited to reviewed labels, stored predictions, group partitions, model configurations, summary statistics, and supporting scripts for numerical verification.

Data category	Protection measure
User identifiers, account names, profile links, source IDs, and URLs	Removed from public artefacts or withheld in restricted internal records
Unredacted comments and OCR/STT text	Reviewed for direct identifiers and released only in de-identified research form where included
Original videos, images, keyframes, and source audio	Excluded from public redistribution and retained only under restricted access where still held
Source-group mappings and internal linkage keys	Maintained separately from public release; not published in the repository
Repository artefacts	Reviewed to avoid publication of direct identifiers, credentials, or unreduced source mappings

The research team recognises that de-identification does not automatically guarantee zero re-identification risk. Accordingly, the repository release was defined conservatively. Materials that could reveal identity through direct disclosure, voice, facial imagery, distinctive text strings, or restricted source mappings were excluded from public release.

7. Release Classification and Repository Controls

Release class	Examples	Control
PUBLIC	Source code, environment files, experiment configurations, de-identified predictions, group partitions, aggregate metrics, statistical outputs, and explanatory documentation	Released after researcher review for privacy and rights consistency
RESTRICTED	Original videos, source audio, identifiable frames, unreduced source text, identity mappings, and selected model checkpoints	Retained separately with controlled access and not distributed through the public repository
EXCLUDED	Credentials, access tokens, unreviewed exports, unnecessary personal data, and any file judged to present unacceptable disclosure risk	Not released; removed or withheld from the public archive

The public repository supports independent verification of stored predictions and reported numerical results. It does not support unrestricted reconstruction of the complete historical collection, original audiovisual corpus, or the entire supervised training pipeline without access to restricted source materials.

8. Operational Reproducibility Boundaries

The manuscript and repository provide sufficient artefacts to verify major numerical claims from stored predictions and documented partitions. The release includes pseudonymised research records, evaluation scripts, statistical outputs, and release inventories. The repository version used for the final public package passed checksum validation, numerical verification, and notebook-based review procedures. End-to-end training reconstruction remains limited by the restriction of original media and selected checkpoints.

Within the aligned video cohort, 450 source windows were excluded after reconciliation checks. Their positive-label proportion differed from that of the retained windows. This exclusion is documented because it forms part of the evidentiary boundary of the released cohort. The public release therefore supports transparent re-evaluation of the reported results, while preserving the restriction of materials that may disclose identities or exceed the release scope.

9. Administrative Declarations

The undersigned Principal Researcher declares that this document accurately reflects the research data governance practices, corpus composition, audit information, and public-release boundaries associated with the reported study, to the extent documented in the research record and the final manuscript. The Head of the S1 Informatics Engineering Study Program acknowledges receipt and administrative review

of this record. Such acknowledgment is not an institutional ethics approval, an ethics-exemption determination, or permission granted by a social-media platform.

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Date: September 23, 2026

Appendix. File-level Release Summary

The following artefact classes were represented in the final research package and repository documentation: (1) de-identified data records and labels, (2) stored out-of-fold predictions and thresholds, (3) source-group partitions and manifests, (4) statistical analysis tables and diagnostic outputs, (5) explanatory notebooks and scripts, and (6) figure assets used to reproduce the reported presentation. Original media and any potentially identifying source mappings were excluded from public release.

REPOSITORY CHRONOLOGY NOTE

Reference prepared on 29 October 2026 using the repository files README.md, RELEASE_INFO.json and CHANGELOG.md (v1.1.2). These are repository milestones, not the institutional issue or signature dates of the accompanying record.

Milestone	Repository date / record
Project release (researcher-designated)	6 October 2026
Analysis reference (researcher-designated)	22 October 2026
Final archive assembly (v1.1.2)	28 October 2026
Institutional record issue and signatures (unaltered)	23 September 2026

The signed administrative record dated 23 September 2026 is retained without alteration. This supplementary chronology documents later repository activity only; it does not assert that the later archive was approved or signed in September, or that an institutional reissuance has occurred.